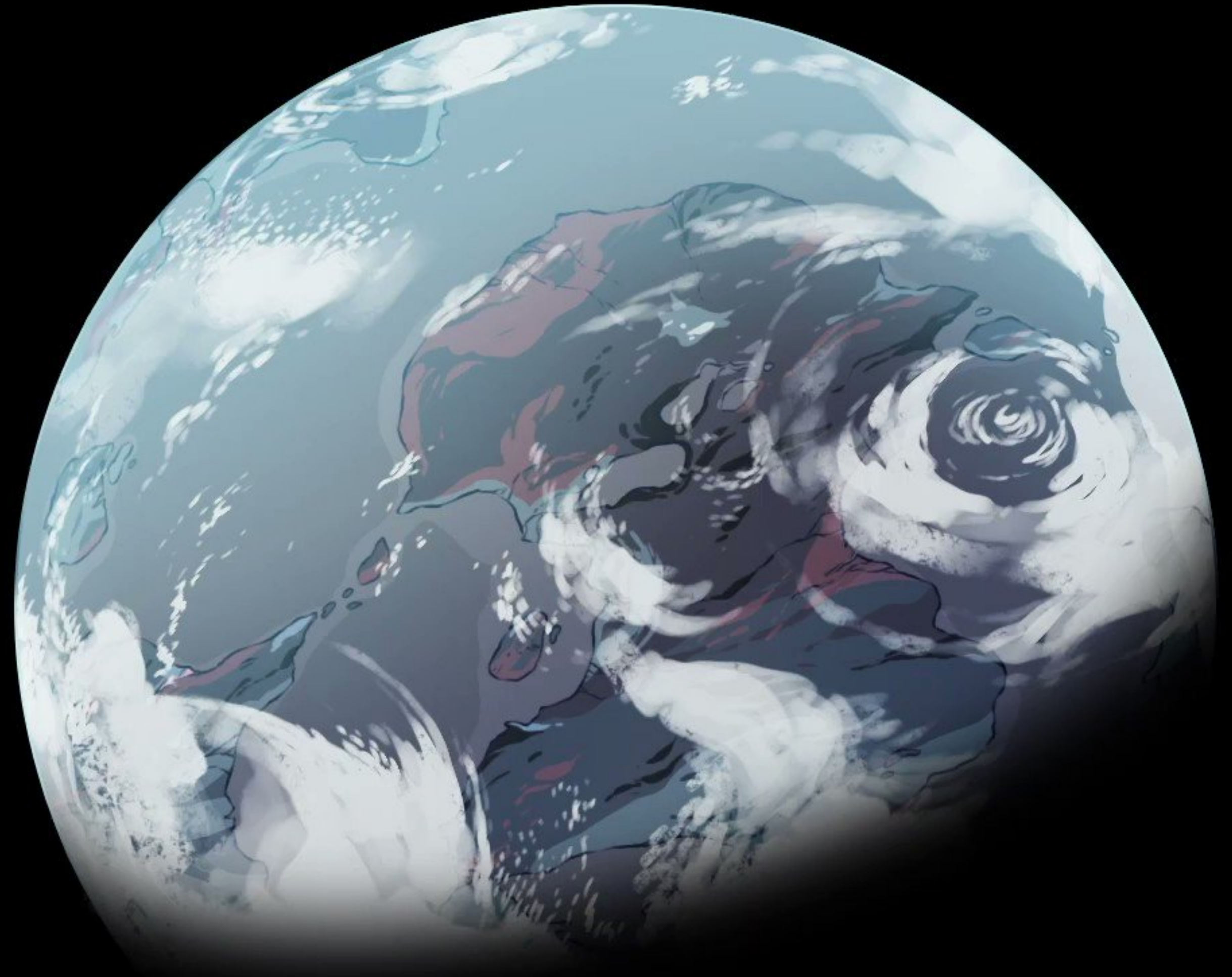


Science & Science Fiction

33-120 | Prof. Katie Breivik



Announcements

- Problem 3 is due tonight at midnight!
- Quiz 2 will be available on Wednesday
 - Covers chapter 2: What is the nature of space and time?

Chapter 3: 09/23 - 10/11



What is the universe made of?

matter

energy

interactions

- ▶ Standard Model of particle physics
- ▶ Dark matter and dark energy
- ▶ Nuclei, isotopes, and radio activity
- ▶ Gases, liquids, and solids
- ▶ Phase transitions
- ▶ Transparency and Invisibility
- ▶ Energy and power

Quantum mechanics: the physics of the tiny

So far we have used two physical constants that describe how space and time operate in our Universe:

$$c = 299,792,458 \text{ m/s}$$

speed of light

sets the speed limit
of the Universe

$$G = 6.6743 \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$$

gravitational constant

sets how strongly space
curves due to mass

Quantum mechanics: the physics of the tiny

If a photon contains energy: E
but has no mass, we can't use
Einstein's rest mass energy,

$$E = mc^2, \text{ describe it}$$

The energy of a photon is
connected to the frequency of
light it emits as $E = hf$

h : Planck constant



Max Planck

Quantum mechanics: the physics of the tiny

The Planck constant describes how energy is *quantized* in our Universe — the “quanta” in quantum mechanics

$$c = 299,792,458 \text{ m/s} \quad G = 6.6743 \times 10^{-11} \text{ m}^3/(\text{kg s}^2) \quad h = 6.626 \times 10^{-34} \text{ J/Hz}$$

speed of light

gravitational constant

Planck constant

sets the speed limit
of the Universe

sets how strongly space
curves due to mass

sets how energy is
quantized

Quantum mechanics: the physics of the tiny

Planck length:

$[l_{\text{Planck}}]$

$$l_{\text{Planck}} = \sqrt{\frac{hG}{2\pi c^3}}$$

$$= \sqrt{\frac{[\text{kg m}^2/\text{s}][\text{m}^3/(\text{kg s}^2)]}{[\text{m}^3/\text{s}^3]}}$$

A useful scaling for
understanding the
smallest scales in our
Universe

$$= \sqrt{[\text{m}^2]} = [m]$$

* experimentalists

* theorists



1927 Solvay conference: electrons and protons

Emmy Noether



Noether's theorem (1915):
systems with symmetry
imply conservation laws

Symmetry in time:

—> conservation of energy

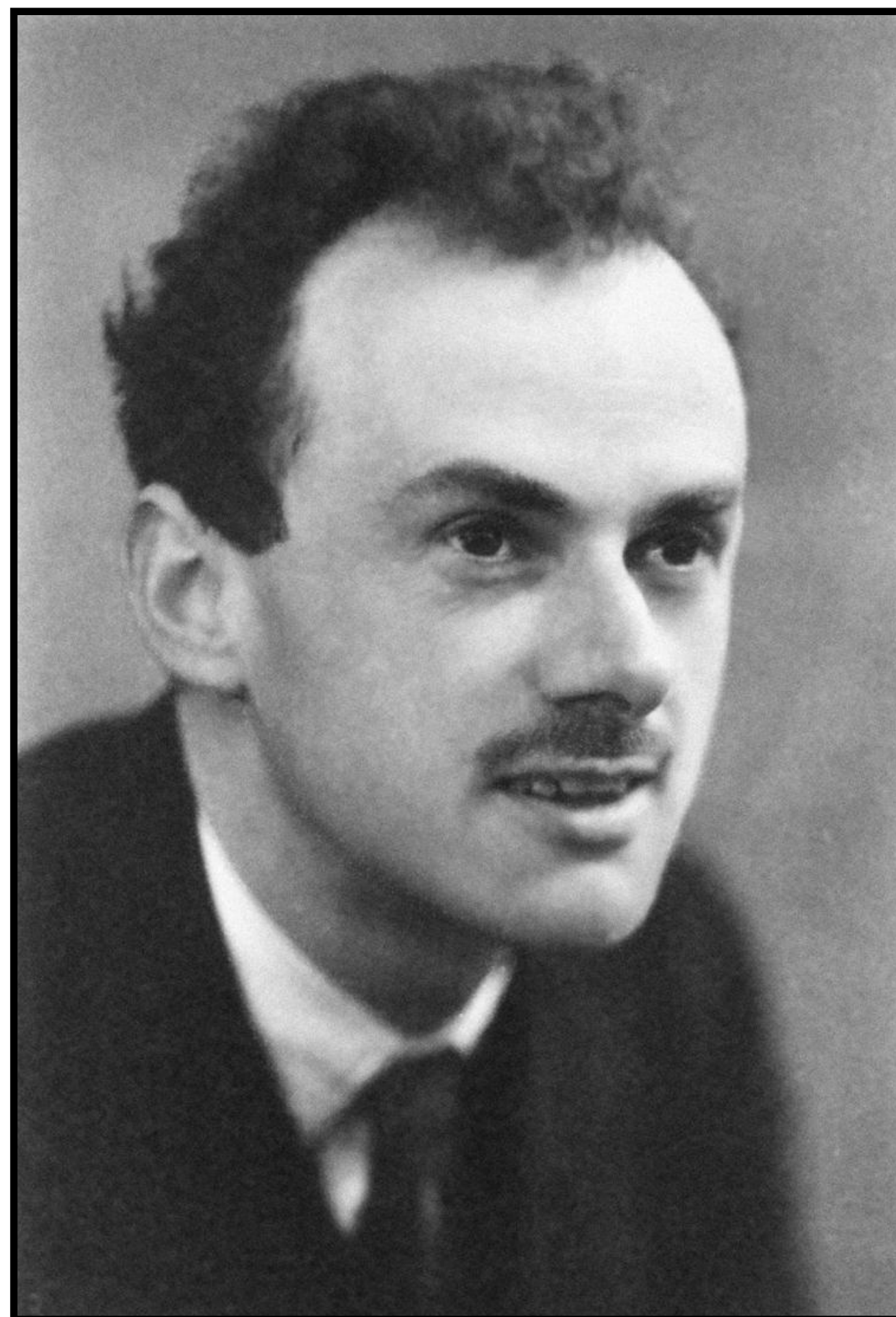
Symmetry in position:

—> conservation of momentum

Symmetry in rotation:

—> conservation of *angular*
momentum

Describes *all* of classical physics!



Paul Dirac

Dirac equation (1928):

$$\left(\beta mc^2 + c \sum_{n=1}^3 \alpha_n p_n \right) \psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

an equation that connects
electromagnetism, quantum mechanics,
and **special relativity** to describe how
electrons work in a relativistic context

Symmetry in *parity* ($\pm$):

—> conservation of **charge**

Standard Model of Elementary Particles

three generations of matter (fermions)				interactions / force carriers (bosons)	
	I	II	III		
QUARKS	mass charge spin $\approx 2.16 \text{ MeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$ u up	$\approx 1.273 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$ c charm	$\approx 172.57 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$ t top	0 0 1 g gluon	$\approx 125.2 \text{ GeV}/c^2$ 0 0 0 H higgs
	$\approx 4.7 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ d down	$\approx 93.5 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ s strange	$\approx 4.183 \text{ GeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ b bottom	0 0 1 γ photon	
	$\approx 0.511 \text{ MeV}/c^2$ -1 $\frac{1}{2}$ e electron	$\approx 105.66 \text{ MeV}/c^2$ -1 $\frac{1}{2}$ μ muon	$\approx 1.77693 \text{ GeV}/c^2$ -1 $\frac{1}{2}$ τ tau	$\approx 91.188 \text{ GeV}/c^2$ 0 1 Z Z boson	
LEPTONS	$< 0.8 \text{ eV}/c^2$ 0 $\frac{1}{2}$ ν_e electron neutrino	$< 0.17 \text{ MeV}/c^2$ 0 $\frac{1}{2}$ ν_μ muon neutrino	$< 18.2 \text{ MeV}/c^2$ 0 $\frac{1}{2}$ ν_τ tau neutrino	$\approx 80.3692 \text{ GeV}/c^2$ ± 1 1 W W boson	

developed by
theorists and
experimentalists
from the 1950s to
the early 2000s

describes what the
Universe is made of



Star Trek:
The Next Generation
“Starship Mine”

Written:
Morgan Gendle

Paramount (1993)

The standard model of particle physics

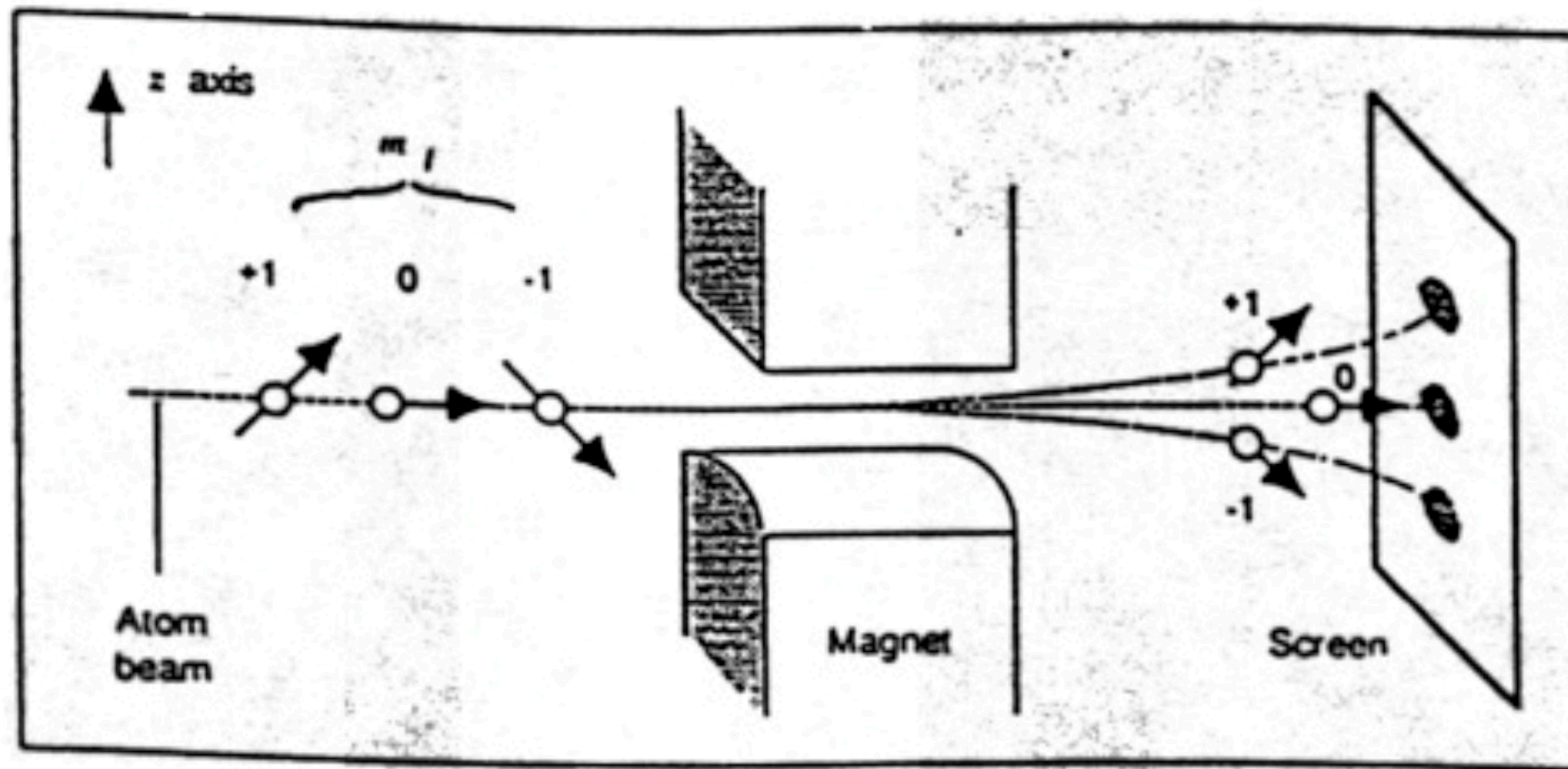
fermions

- ▶ Fundamental particles of matter
- ▶ Quantum spin:
 $\pm 1/2, \pm 3/2, \pm 5/2$, etc.

bosons

- ▶ Fundamental force communicators
- ▶ Quantum spin:
0, 1, 2, etc.

What is quantum spin???



the Stern-Gerlach experiment (1922)

fermions: fundamental particles of *matter*

quarks

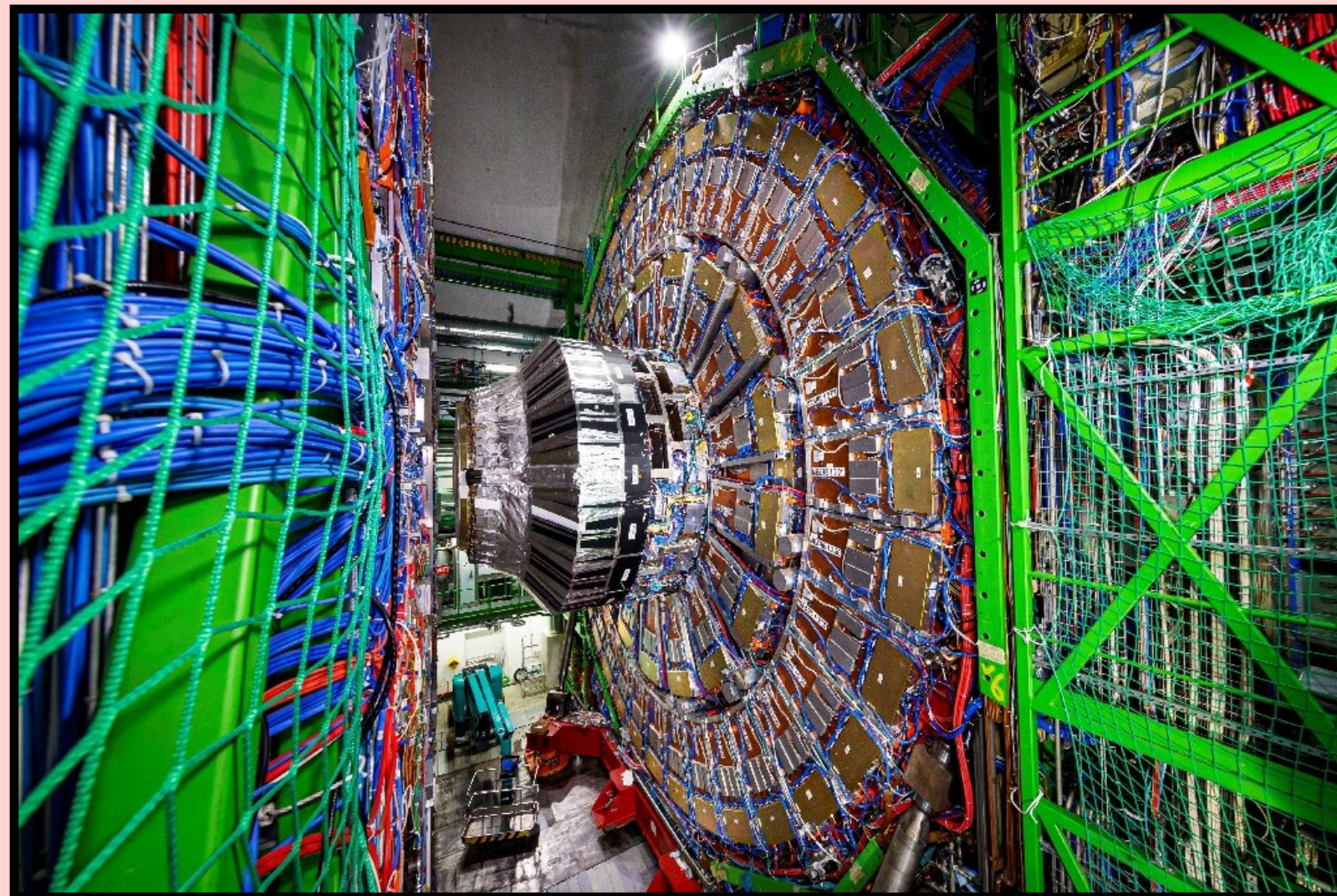
- ▶ up, down
- ▶ charm, strange
- ▶ top, bottom

leptons

- ▶ electron, e-neutrino
- ▶ muon, mu-neutrino
- ▶ tau, tau-neutrino

Most stuff is made of fermions

Hadrons: things made up of quarks



The large hadron collider (LHC) was built at CERN to crash hadrons together to study particle physics

Most stuff is made of fermions

Hadrons: things made up of quarks

mesons:
quark &
anti-quark
pairs

baryons:
things made
of three
quarks



Would the ship
survive a baryon
sweep?

Nope!

bosons: fundamental *force* communicators

fundamental force

boson

electromagnetism

photon

weak nuclear force

(radioactive decay)

$W^{\pm}(\text{eak}), Z(\text{ero})$

strong nuclear force

(binds nucleons together)

gluon

gravity

graviton?

bosons: fundamental *force* communicators

fundamental force

boson

electromagnetism

photon*

weak nuclear force
(radioactive decay)

massless*

$W^{\pm}(\text{eak}), Z(\text{ero})^*$

strong nuclear force
(binds nucleons together)

massive*

gluon*

gravity

graviton?*

Standard Model of Elementary Particles

three generations of matter (fermions)				interactions / force carriers (bosons)	
	I	II	III		
mass	$\approx 2.16 \text{ MeV}/c^2$	$\approx 1.273 \text{ GeV}/c^2$	$\approx 172.57 \text{ GeV}/c^2$	0	$\approx 125.2 \text{ GeV}/c^2$
charge	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	0	0
spin	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	0
QUARKS	u up	c charm	t top	g gluon	H higgs
	$\approx 4.7 \text{ MeV}/c^2$	$\approx 93.5 \text{ MeV}/c^2$	$\approx 4.183 \text{ GeV}/c^2$	0	
	$-\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	0	
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	
	d down	s strange	b bottom	γ photon	
LEPTONS	$\approx 0.511 \text{ MeV}/c^2$	$\approx 105.66 \text{ MeV}/c^2$	$\approx 1.77693 \text{ GeV}/c^2$	$\approx 91.188 \text{ GeV}/c^2$	
	-1	-1	-1	0	
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	
	e electron	μ muon	τ tau	Z Z boson	
	$< 0.8 \text{ eV}/c^2$	$< 0.17 \text{ MeV}/c^2$	$< 18.2 \text{ MeV}/c^2$	$\approx 80.3692 \text{ GeV}/c^2$	
	0	0	0	± 1	
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson	



Peter Higgs
(1964)

Higgs boson mediates the **Higgs field** which breaks the symmetry of the weak nuclear force

Explains how the Z, W bosons (and all other particles) have mass

Discovery announced in 2013 @ LHC

Next time:

▶ Dark matter, Dark energy

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